# Stage-2 Anchor Ablation
# LONO Validation of the Early-Time Physics Prior

Mie Spray Postprocessing Project

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# Message

**Claim:** adding a soft $\boldsymbol{\mu(t\to 0)\approx 0}$ anchor to the Stage-2 NLL loss improves out-of-nozzle generalisation. Extending the anchor to also constrain $\sigma$ brings no further gain (and slightly over-tightens uncertainty).

**This deck:**

- recaps the physical motivation: penetration must start at $0$ at injection onset, so $\mu$ and $\sigma$ near $t{=}0$ should be small
- defines the three anchor ablations: `no_anchor`, `mu_anchor`, `mu_sigma_anchor`
- shows per-fold and mean post-Stage-3 RMSE/MAE/coverage for all three under 5-fold LONO
- confirms (i) `mu_anchor` wins on RMSE, MAE, and $2\sigma$ coverage, (ii) `no_anchor` catastrophically fails on the hardest non-Nozzle0 fold, (iii) adding $\sigma$-anchor narrows predicted $\sigma$ but does not improve RMSE

**Validated pipeline:** `run_stage2_lono_ablation.py`, seed 42, 5 folds $\times$ 3 ablations (uncensored eval points per fold $\in \{311149, 60369, 33017, 28139, 25705\}$); Stage-1 backbone fixed to `a_dp050_plus_pressures` (Stage-1 winner).

**Physical constraint at injection onset:**

$$S(t{=}0) = 0, \qquad \sigma\big(S(t{=}0)\big) \text{ should be small.}$$

Without an explicit penalty the NLL trains the network to optimise the *likelihood of the observed points*, which contain almost no samples at $t < 0.1\,\mathrm{ms}$ (laser onset truncation, shadow latency, etc.). The network therefore drifts to large $|\mu|$ and inflated $\sigma$ near $t{=}0$ just because that region is data-poor.

**Anchor penalty** (added to the NLL):

$$\mathcal{R}_\mu = \frac{\sum_t w(t)\,\mu(t)^2}{\sum_t w(t)}, \qquad \mathcal{R}_\sigma = \frac{\sum_t w(t)\,\big[\max(0,\,\sigma(t) - \sigma_0)\big]^2}{\sum_t w(t)},$$

$$w(t) = \max\!\Big(0,\ 1 - \tfrac{t}{\tau}\Big), \qquad \tau = \text{anchor\_window\_ms (early-time ramp)}.$$

$w(t)$ is a triangular kernel: weight $1$ at $t{=}0$, decaying linearly to $0$ at $t{=}\tau$.

**Final Stage-2 loss:**

$$\mathcal{L}_2 \;=\; \mathrm{NLL}\big(y;\,\mu(x),\sigma(x)\big) \;+\; \lambda_\mu\,\mathcal{R}_\mu \;+\; \lambda_\sigma\,\mathcal{R}_\sigma.$$

# Variant Design: What Each Configuration Tests

| Variant | $\lambda_\mu$ | $\lambda_\sigma$ | What it tests |
|---|---|---|---|
| `no_anchor` | 0 | 0 | pure NLL baseline; no physics prior at $t{=}0$ |
| `mu_anchor` | $10^{-2}$ | 0 | soft anchor only on $\mu$ —penalises non-zero penetration at injection onset |
| `mu_sigma_anchor` | $10^{-2}$ | $10^{-3}$ | adds soft anchor on excess-$\sigma$ above the floor $\sigma_0$ |

**Identical across ablations:**

- Stage-1 backbone: `a_dp050_plus_pressures` (locked from Stage-1 LONO winner)
- Stage-3 chain: `anchor_off` distillation (identical recipe)
- Same train/val split, same seed (42), same nozzle held out per fold

**Per-fold pipeline:** Stage-1 (1 run) $\rightarrow$ Stage-2 (3 ablations) $\rightarrow$ Stage-3 anchor_off (3 chains) $\rightarrow$ post-Stage-3 RMSE eval on the held-out nozzle's uncensored points.

Total trainings: $5 \times (1{+}3{+}3) = 35$ runs; reported metrics are the post-Stage-3 eval (end-task quality).

# LONO Results: Summary (Mean $\pm$ Std Across 5 Folds)

| Ablation | RMSE (mm) | MAE (mm) | Bias (mm) | cov$_{2\sigma}$ | $\bar{\sigma}$ (mm) |
|---|---|---|---|---|---|
| mu_anchor | **12.73 ± 12.44** | **9.86 ± 10.22** | $-7.11 \pm 11.51$ | **0.775 ± 0.34** | 5.68 ± 1.44 |
| mu_sigma_anchor | 13.05 ± 12.46 | 10.06 ± 10.33 | $-7.59 \pm 11.45$ | 0.764 ± 0.35 | **5.49 ± 1.55** |
| no_anchor | 14.54 ± 12.36 | 10.80 ± 9.96 | $-8.50 \pm 11.03$ | 0.763 ± 0.34 | 6.11 ± 1.58 |

**Headline numbers, excluding the outlier Nozzle0 fold (4-fold mean RMSE):**

| Ablation | RMSE w/o Nozzle0 (mm) |
|---|---|
| mu_anchor | **7.18** |
| mu_sigma_anchor | 7.53 |
| no_anchor | 9.61 |

Green: anchored variants. Red: no-anchor baseline. `mu_anchor` wins on RMSE, MAE, and $2\sigma$-coverage; `mu_sigma_anchor` has the tightest predicted $\sigma$ (5.49 mm) but loses 0.32 mm RMSE —over-constraining the uncertainty channel hurts the mean.

# LONO Results: Per-Fold Breakdown

| Ablation | Nozzle0 (311149) | Nozzle2 (60369) | Nozzle5 (33017) | Nozzle4 (28139) | Nozzle1 (25705) | Mean |
|---|---|---|---|---|---|---|
| *RMSE (mm)* | | | | | | |
| mu_anchor | 34.92 | 7.76 | 5.60 | 8.20 | 7.16 | **12.73** |
| mu_sigma_anchor | 35.12 | 10.04 | 5.27 | 7.73 | 7.10 | 13.05 |
| no_anchor | 34.25 | 19.18 | 5.11 | 7.57 | 6.56 | 14.54 |
| *MAE (mm)* | | | | | | |
| mu_anchor | 28.11 | 5.72 | 4.26 | 5.84 | 5.37 | 9.86 |
| mu_sigma_anchor | 28.44 | 6.91 | 3.89 | 5.79 | 5.29 | 10.06 |
| no_anchor | 27.53 | 12.46 | 3.79 | 5.41 | 4.82 | 10.80 |
| *Coverage at $2\sigma$ (target 0.95)* | | | | | | |
| mu_anchor | .171 | .856 | .975 | .912 | .963 | .775 |
| mu_sigma_anchor | .149 | .801 | .981 | .929 | .959 | .764 |
| no_anchor | .191 | .690 | .985 | .965 | .982 | .763 |

Post-Stage-3 metrics on held-out nozzle's uncensored CDF P50 points. Number under each nozzle = eval-point count. Columns sorted by descending point count; Nozzle0 dominates the mean by sample weight.

# Discussion: Three Consistent Patterns

## 1. $\mu$-anchor stabilises the hardest non-outlier fold

On Nozzle2 (the second-largest fold by sample count and the only fold where the model is asked to extrapolate to a new injector geometry within the modern family), `no_anchor` collapses to 19.18 mm RMSE while `mu_anchor` holds at 7.76 mm. The anchor prevents the NLL from fitting an arbitrary near-$t$=0 baseline using the data-poor early window as free parameters.

## 2. Adding $\sigma$-anchor doesn't recover the lost RMSE

`mu_sigma_anchor` achieves the tightest predicted $\sigma$ (5.49 mm mean) but its RMSE is 0.32 mm worse than `mu_anchor`. Pushing $\sigma$ down at $t$=0 propagates through Stage-3 distillation as a more confident-but-biased $\mu$ prediction, hurting accuracy without improving calibration ($\text{cov}_{2\sigma}$ 0.764 vs 0.775).

## 3. Easy folds (Nozzle1/4/5) are insensitive to the anchor choice

On the three modern-design nozzles with ∼25–33k eval points, all three ablations sit in a narrow 5.1–8.2 mm RMSE band. The anchor only matters when the LONO fold creates a genuine OOD challenge (Nozzle0 for geometric reasons, Nozzle2 for sample-size reasons).

# Discussion: Nozzle0 is Still the Hardest Fold

Nozzle0 (`BC20220627_HZ_Nozzle0`) carries **311,149 uncensored eval points** —about 5× more than the next-largest fold (Nozzle2: 60,369). It is also the *only* first-generation nozzle in the dataset.

All three ablations land in the same catastrophic band on this fold:

- RMSE 34–35 mm
- bias $\approx -27$ mm (systematic under-prediction)
- $2\sigma$ coverage 0.15–0.19

**Interpretation:** the Nozzle0 failure is not an anchor problem; it is a fundamental *out-of-design-family* extrapolation. None of the in-distribution Stage-2 regularisers can compensate for a missing geometric class. This was already known from the Stage-1 ablation and is reproduced here at the Stage-3 output level.

| Fold | Points | Best RMSE |
|---|---|---|
| Nozzle0 | 311,149 | 34.25 mm |
| Nozzle2 | 60,369 | 7.76 mm |
| Nozzle5 | 33,017 | 5.11 mm |
| Nozzle4 | 28,139 | 7.57 mm |
| Nozzle1 | 25,705 | 6.56 mm |

Best RMSE = minimum across the three ablations for that fold. Excluding Nozzle0 collapses the cross-ablation gap to 7.18 mm (`mu_anchor`) vs 9.61 mm (`no_anchor`), a 2.4 mm OOD margin from the anchor alone.

# Conclusion: $\mu$-Anchor is the Stage-2 LONO Winner

## What the ablation proves

1. **The early-time $\mu$-anchor is load-bearing.** It buys 1.8 mm mean RMSE (14.54 $\rightarrow$ 12.73) over no-anchor with Nozzle0 included, and 2.4 mm (9.61 $\rightarrow$ 7.18) excluding Nozzle0.
2. **The anchor's value shows up on hard OOD folds, not easy ones.** On Nozzle1/4/5 all variants are within 0.5 mm; on Nozzle2 the anchor saves 11+ mm. The regularisation only matters when the model is forced to extrapolate.
3. **$\sigma$-anchor is not additive.** Tightening $\sigma$ at $t=0$ trades $\bar{\sigma}$ (5.68 $\rightarrow$ 5.49 mm) for a worse RMSE (12.73 $\rightarrow$ 13.05 mm) and slightly worse calibration. `mu_anchor` is the Pareto winner.
4. **Production variant for downstream Stage-3 LONO and multi-seed retrains:** `mu_anchor` ($\lambda_\mu=10^{-2}$, $\lambda_\sigma=0$).

**Experiment:** `run_stage2_lono_ablation.py` — 3 ablations × 5 LONO folds, seed 42, Stage-1 = `a_dp050_plus_pressures`, Stage-3 = `anchor_off`.
**Output:** `MLP/runs_mlp/stage2_lono_ablation_20260520_235200/`